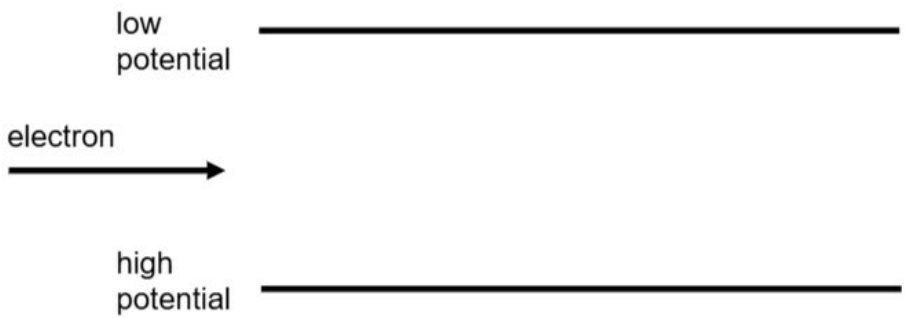
24 An electron is projected horizontally with a speed of 3.1×10^7 m s^{-1}. It passes in between the two charged horizontal plates. The uniform electric field between the plates is 20000 N C^{-1}. As the electron pass through the plates, it is deflected. The time taken for the electron to travel the length of the plate is 4.0×10^{-9} s.



What is the vertical distance travelled by the electron after it has travelled the length of the plate?

A 1.76×10^{-17} m **B** 7.85×10^{-17} m **C** 0.0281 m **D** 0.152 m